



University of Science and Technology

Electronic engineering



Tutorial of Physics II

COURSE COORDINATOR

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Coulomb's Law and Electric Fields

$$F_E = k \frac{qq'}{r^2} \quad (\text{in vacuum})$$

$$k = 8.988 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$$

$K\epsilon_0 = \epsilon$, where ϵ is called the *permittivity of the material*.

$$e = 1.60218 \times 10^{-19} \text{ C.}$$

ELECTRIC FIELD

$$E = \frac{1}{4\pi\epsilon} \frac{q}{r^2}$$

$$F_E = q'E$$

If two equal charges, each of 1 C, were separated in air by a distance of 1 km, what would be the force between them? *Ans.* 9 kN repulsion

solution

$$F = k \frac{q_1 q_2}{r^2}$$

$$q_1 = q_2 = 1, \quad r = 1 \times 10^3 \text{ m}, \quad F = \dots$$

$$F = 9 \times 10^9 \frac{(1^2)}{(10^3)^2} = 9 \times 10^{9-6} = 9 \times 10^3 = 9 \text{ kN}$$

Determine the force between two free electrons spaced 1.0 angstrom (10^{-10} m) apart. *Ans.* 23 nN
repulsion

solution

$$F = k \frac{q_1 q_2}{r^2} = k \frac{e^2}{r^2}$$

$$e^2 = (1.6 \times 10^{-19})^2, \quad r = 1 \times 10^{-10} \text{ m}, \quad F = \dots$$

$$F = 9 \times 10^9 \frac{(1.6 \times 10^{-19})^2}{(10^{-10})^2} = 9 \times 1.6^2 \times 10^{9+20-38} = 23.04 \times 10^{-9}$$

$$= 23.04 \text{ nN} = 23 \text{ nN}$$

What is the force of repulsion between two argon nuclei that are separated by 1.0 nm (10^{-9} m)? The charge on an argon nucleus is $+18e$. *Ans.* 75 nN

solution

$$F = k \frac{q_1 q_2}{r^2}$$

$$q_1 = q_2 = +18e = 18 \times 1.6 \times 10^{-19}), \quad r = 1 \times 10^{-9} \text{ m}, \quad F = \dots$$

$$F = 9 \times 10^9 \frac{(18 \times 1.6 \times 10^{-19})^2}{(10^{-9})^2} = 9 \times (18 \times 1.6)^2 \times 10^{9+18-38}$$

$$= 7464.96 \times 10^{-11} \text{ N} = 74.6496 \times 10^{-9} = 75 \text{ nN}$$

Two equally charged balls are 3 cm apart in air and repel each other with a force of $40\mu\text{N}$. Compute the charge on each ball. *Ans.* 2 nC

solution

$$F = k \frac{q_1 q_2}{r^2} \quad \equiv \quad q^2 = \frac{F \times r^2}{k}$$

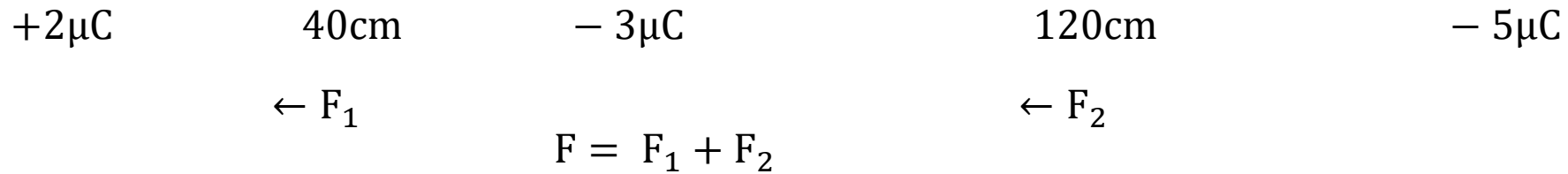
$$q_1 = q_2 = q^2, \quad r = 3 \times 10^{-2} \text{m}, \quad F = 40 \times 10^{-6} \text{ N}$$

$$q^2 = \frac{40 \times 10^{-6} \times 9 \times 10^{-4}}{9 \times 10^9} = 4 \times 10^{-5-4-9} = 4 \times 10^{-18} \text{ C}^2$$

$$q = \sqrt{4 \times 10^{-18}} = 2 \times 10^{-9} = 2 \text{ nC}$$

Three point charges are placed at the following points on the x -axis: $+2.0\ \mu\text{C}$ at $x=0$, $-3.0\ \mu\text{C}$ at $x=40\text{ cm}$, $-5.0\ \mu\text{C}$ at $x=120\text{ cm}$. Find the force (a) on the $-3.0\ \mu\text{C}$ charge, (b) on the $-5.0\ \mu\text{C}$ charge. *Ans. (a) -0.55 N ; (b) 0.15 N*

solution



$$F_1 = 9 \times 10^9 \frac{2 \times 10^{-6} \times 3 \times 10^{-6}}{(4 \times 10^{-1})^2} = \frac{9 \times 2 \times 3}{16} \times 10^{-12+9+2} = 3.375 \times 10^{-1} = 0.34\text{ N}$$

$$F_2 = 9 \times 10^9 \frac{5 \times 10^{-6} \times 3 \times 10^{-6}}{(8 \times 10^{-1})^2} = \frac{9 \times 5 \times 3}{64} \times 10^{-12+9+2} = 2.1094 \times 10^{-1} = 0.21\text{ N}$$

$$\mathbf{F = 0.34 + 0.21 = 0.55\text{ N}}$$

— direction of force — Xaxis

Three point charges are placed at the following points on the x -axis: $+2.0\ \mu\text{C}$ at $x = 0$, $-3.0\ \mu\text{C}$ at $x = 40\text{ cm}$, $-5.0\ \mu\text{C}$ at $x = 120\text{ cm}$. Find the force (a) on the $-3.0\ \mu\text{C}$ charge, (b) on the $-5.0\ \mu\text{C}$ charge. *Ans.* (a) -0.55 N ; (b) 0.15 N

solution



$$F = F_1 - F_2 \text{ or } F_2 - F_1$$

$$F_1 = 9 \times 10^9 \frac{2 \times 10^{-6} \times 5 \times 10^{-6}}{(12 \times 10^{-1})^2} = \frac{9 \times 2 \times 5}{144} \times 10^{-12+9+2} = 0.625 \times 10^{-1} = 0.06 \text{ N}$$

$$F_2 = 9 \times 10^9 \frac{5 \times 10^{-6} \times 3 \times 10^{-6}}{(8 \times 10^{-1})^2} = \frac{9 \times 5 \times 3}{64} \times 10^{-12+9+2} = 2.10 \text{WE} \times 10^{-1} = 0.21 \text{ N}$$

$$F = 0.21 - 0.06 = 0.15 \text{ N}$$

A point charge of $-3.0 \times 10^{-5} \text{ C}$ is placed at the origin of coordinates. Find the electric field at the point $x = 5.0 \text{ m}$ on the x -axis. *Ans.* 11 kN/C in $-x$ -direction

solution

$$E = k \frac{q}{r^2}$$

$$\begin{aligned} E &= k \frac{q}{r^2} = 9 \times 10^9 \frac{3 \times 10^{-5}}{5^2} = \frac{9 \times 3}{25} \times 10^{9-5} = 1.08 \times 10^4 \text{ N/C} \\ &= 11 \text{ kN/C} \end{aligned}$$

1. The direction of electric field created by a negative charge is _____

a) Directed outwards

b) Directed towards the charge

c) Maybe outwards or towards the charge

d) Circular in shape

2. V/m is the unit of _____

a) Electric field intensity

b) Electric flux

c) Electric potential

d) Charge

3. Two electric field lines _____

a) Always intersect each other

b) Never intersect

c) May intersect sometimes

d) Are always perpendicular to each other

4. Coulomb is the unit of which quantity?

a) Field strength

b) Charge

c) Permittivity

d) Force

5. Which among the following is false about electric field lines?

- a) They are continuous
- b) They attract each other
- c) They remain parallel in a uniform electric field
- d) They diverge from positive charge

6. The amount of force exerted on a unit positive charge in an electric field is known as ____

- a) Electric field intensity
- b) Electric flux
- c) Electric potential
- d) Electric lines of force

7. Find the force between 2C and -1C separated by a distance 1m in air(in newton).

- a) 18×10^6
- b) 18×10^9
- c) 18×10^{-6}
- d) 18×10^8

8. Coulomb law is employed in

- a) Electrostatics
- b) Magnetostatics
- c) Electromagnetics
- d) Maxwell theory

9. Two charges $1C$ and $-4C$ exist in air. What is the direction of force?

- a) Away from $1C$
- b) Away from $-4C$
- c) From $1C$ to $-4C$
- d) From $-4C$ to $1C$

10. The amount of force exerted on a unit positive charge in an electric field is known as _____

- a) Electric field intensity
- b) Electric flux
- c) Electric potential
- d) Electric lines of force

11. The direction of electric field created by a negative charge is _____

- a) Directed outwards
- b) Directed towards the charge
- c) Maybe outwards or towards the charge
- d) Circular in shape

12. Electric field is a _____

- a) Scalar quantity
- b) Vector quantity
- c) Tensor quantity
- d) Quantity that has properties of both scalar and vector

13. What is the dimension of electric field intensity?

- a) $[M L T^{-2} I^{-1}]$
- b) $[M L T^{-3} I^{-1}]$
- c) $[M L T^{-2} I^{-2}]$
- d) $[M L T^{-3} I]$

14. Find the electric field intensity at 3 cm away from a point charge of 100 n C?

- a) 1 N/C
- b) 10 N/C
- c) 100 N/C
- d) 9×10^9 N/C

15. The [electric](#) field intensity is defined as

- a) Force per unit charge
- b) Force on a test charge
- c) Force per unit charge on a test charge
- d) Product of force and charge

16. Find the force on a charge 2C in a field 1V/m.

- a) 0
- b) 1
- c) 2
- d) 3

17. Find the electric field intensity of two charges 2C and -1C separated by a distance 1m in air.

- a) 18×10^9
- b) 9×10^9
- c) 36×10^9
- d) -18×10^9

18. What is the electric field intensity at a distance of 20cm from a charge 2×10^{-6} C in vacuum?

- a) 250,000
- b) 350,000
- c) 450,000
- d) 550,000

19. Determine the charge that produces an [electric field strength](#) of 40 V/cm at a distance of 30cm in vacuum (in 10^{-8} C)

- a) 4
- b) 2
- c) 8
- d) 6

20. For a test charge placed at infinity, the electric field will be

- a) Unity
- b) $+\infty$
- c) Zero
- d) $-\infty$